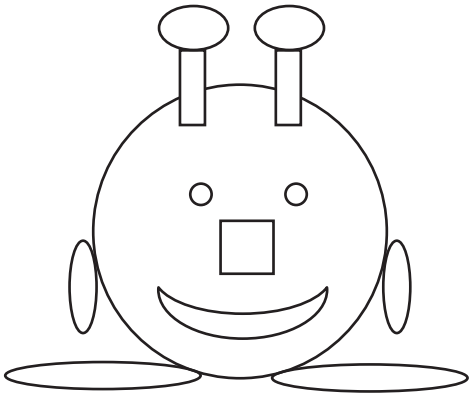
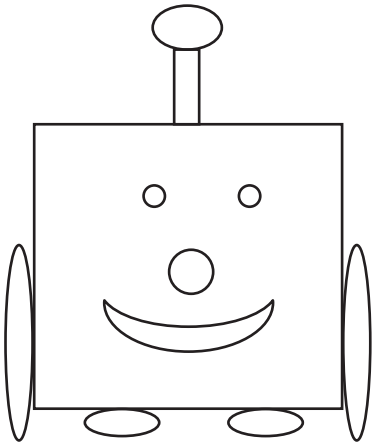
Genetics with the Beeple people

13a

The Beeple people from the planet Genetoz have volunteered to show you their chromosomes to help you develop a clearer understanding of how inheritance works.

Pictured below are Mum and Dad Beeple. Mum is expecting a baby. By using Mum and Dad's characterictics, you will be able to predict what their baby may look like.





Describing Mum's and Dad's looks: Phenotype

Mum Beeple has the most amazing brown eyes, a cute orange nose and elegant long blue feet. What attracted her to Dad was his bright green eyes. He too has an orange nose, but his best feature is his short red feet.

1 Colour Mum and Dad to show what they look like – that is, their phenotypes.

Describing Mum's and Dad's genes: Genotype

Look at the characteristics decoder table at the right. Remember, every characteristic is determined by two genes. The dominant gene, shown by a capital letter, usually 'covers up' the recessive gene, which is shown by a small letter.

2 Use the characteristics decoder table to find out which gene pair (genotype) matches each appearance (phenotype). Complete the genotype table below, which shows the genes Mum and Dad have for each trait. Some genes have been included for you.

Characteristics decoder table

Trait	Appearance (phenotype)	Gene pair (genotype)
Sex	Male: square Female: round	XY XX
Eye colour	Brown Green	BB or Bb bb
Nose shape	Square Round	SS or Ss ss
Nose colour	Red Orange Yellow	RR Rr rr
Feet length	Long Short	LL or Ll ll
Feet colour	Purple Red Blue	PP Pp pp
Arm length	Long Short	AA or Aa aa
Number of antennae	One antenna Two antennae Three antennae	NN Nn nn

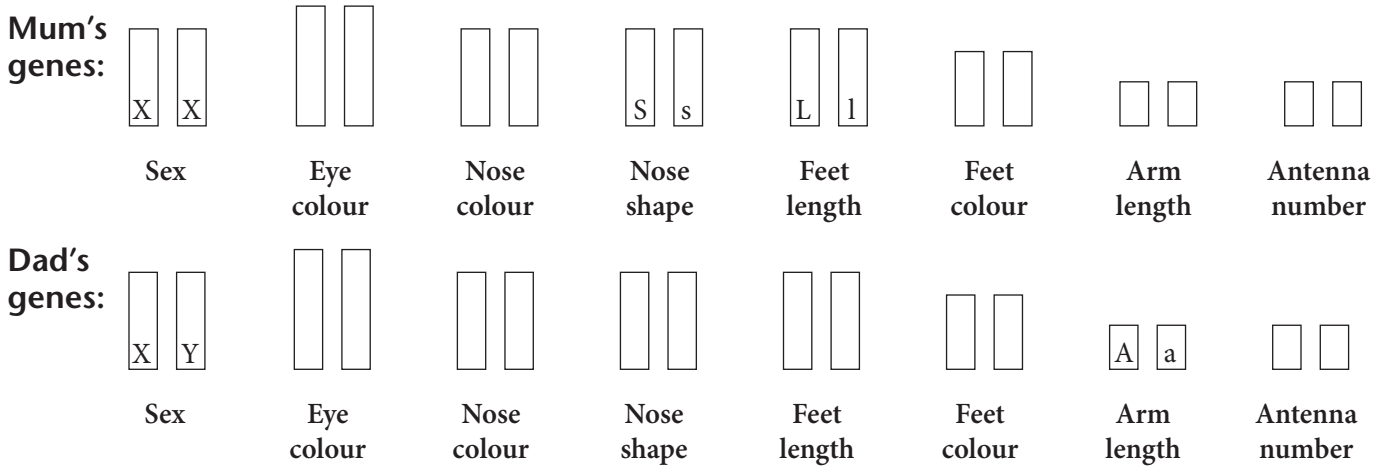
	Trait							
	Sex	Eye colour	Nose colour	Nose shape	Feet length	Feet colour	Arm length	Number of antennae
Mum		BB		Ss	Ll			
Dad							Aa	

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Genetics with the Beeple people

Describing Mum's and Dad's chromosomes: Karyotyping

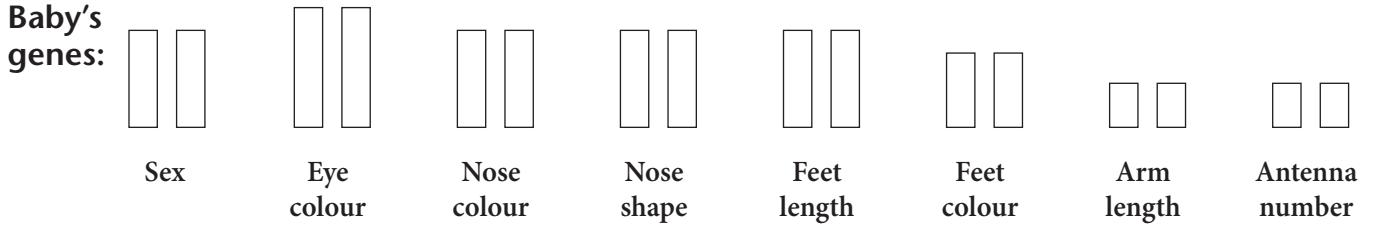
3 The chromosomes for Mum and Dad have been sorted into matching pairs below. Each gene for a characteristic occurs on a different chromosome. Write one letter onto each chromosome to show which gene is present. Some genes have been included for you.



Making and describing Baby

4 It's time to predict what Mum and Dad's baby may look like. Follow the steps below.

- Work in pairs, with one person representing Mum and the other person representing Dad.
- Look at the first chromosome pair from Mum's and Dad's gene charts above. Each person should flip a coin to choose one gene at random from their pair. Record the letter of the gene in Baby's chart below.
- Repeat this procedure, choosing chromosomes at random from the other seven chromosome pairs.



5 In the table below, show Baby's genotype by referring to the chromosome pairs above. Then refer to the characteristics decoder table on the previous page to identify Baby's phenotype.

Baby's traits	Sex	Eye colour	Nose colour	Nose shape	Feet length	Feet colour	Arm length	Number of antennae
Genotype								
Phenotype								

6 Use the illustrations of Mum and Dad Beeple on the previous page to draw and colour a picture of Baby.

7 How many chromosomes:

a do Mum and Dad have?_____ **b** does Baby have?_____

8 Does your baby look like Mum or Dad?_____

9 Compare babies with the students next to you. Are they similar?_____

10 Does anyone in your class have a baby:

a identical to yours?_____ **b** exactly the opposite of yours?_____

11 **Challenge!** Work in small groups to figure out how many possible gene combinations there are for Baby.

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